

Data Challenge

Role

Labour Market Analyst at Global Think Tank.

Role of Company (Global Think Tank)

Advising governments and employers on the workforce impact of artificial intelligence.

Questions

The Director has asked you to investigate how AI adoption is reshaping employment across countries, industries, and skill categories between 2021 and 2024, with particular attention to the shift since the arrival of generative AI tools in late 2022.

The Scenario:

Your think tank has been commissioned by an international policy coalition to produce briefing on AI-driven workforce displacement and reskilling needs. Member governments want to understand which industries and skill categories are most exposed to AI-driven change, how adoption rates have shifted since generative AI tools became mainstream, and where reskilling investment is failing to keep pace with job displacement.

Your findings will directly inform funding allocations for national reskilling programmes due to be announced later this year.

Dataset Overview

Dimensions

4 dimension tables

1. Country
2. Industry
3. Skill category
4. Date

Facts

1 Fact table recording quarterly AI adoption and workforce displacement metrics

Scope

Spans 2021-Q1 to 2024-Q4 (16 Quarters)

Rows

300 fact records across 30 countries, 25 industries and 8 skill categories

Data is synthetic

Key Business Metrics

- `ai_adoption_rate` – percentage of the workforce in a given country/industry/skill segment actively using AI tools
- `displacement_risk_index` – composite 0-10 score estimating a segment's exposure to AI driven job displacement
- `jobs_displace_count / jobs_created_count` – net employment shift attributable to AI adoption
- `reskilling_investment_usd` – capital invested in workforce reskilling for the segment
- `data_confidence_score` – a 0-1 measure of how reliable the underlying survey data is for that record

Analysis Areas

Look for patterns and insights across these key areas:

- How AI adoption rates differ before and after the generative AI era (post-Q4 2022) across industries
- Which skill categories carry the highest displacement risk, and whether reskilling investment is proportional to that risk
- Whether developed and emerging economies are adopting AI at different rates, and what that implies for the global jobs gap
- Where job creation is offsetting (or failing to offset) job displacement at a country or industry level.

Guiding Trends

Use these questions to guide your analysis. They are starting points – feel free to explore further.

Temporal Trends

- How has `ai_adoption_rate` changed quarter over quarter since the generative AI era began in Q4 2022?
- Is the rate of job displacement accelerating or stabilizing across 2023-2024?

Industry and Skill Analysis

- Which industries combine high `automation_susceptibility` with low `avg_ai_investment_pct_revenue`?
- Which skills categories show the largest gap between `ai_replaceability_score` and `median_reskilling_duration_months`?
- Are creative and interpersonal skill categories more resilient to displacement than administrative or manual ones?

Country-Level Patterns

- How does `displacement_risk_index` correlate with `digital_infrastructure_score` and `stem_graduates_per_100k` by country?
- Do countries with more mature `ai_policy_maturity` show better job creation-to-displacement ratios?
- Which regions show the widest gap between developed and emerging economy adoption rates?

Reskilling and Investment

- Is `reskilling_investment_usd` keeping pace with `jobs_displaced_count` at an industry level?
- Which combination of country and industry shows the largest shortfall between displacement risk and reskilling spend?

Impact & Purpose

Your analysis should help policy stakeholders:

- Identify which countries, industries, and skill groups need urgent reskilling investment
- Understanding how the generative AI era has changed the pace of workforce displacement
- Make the case for where coalition funding should be targeted
- Anticipate which segments are likely to face the sharpest disruption over the next two years

Deliverables

Create a professional analytical report that includes:

1. Executive Summary
 - a. Key findings and recommendations
 - b. High-impact insights in plain language
2. Detailed Analysis
 - a. Visualizations showing trends and patterns across time, industry and geography
 - b. Statistical summaries and breakdowns by skill category and development tier
 - c. Answers to the guiding questions
3. Actionable Insights
 - a. What you discovered and why it matters for policy decisions
 - b. Recommend next steps for reskilling investment targeting
 - c. Areas for deeper investigation

Data Dictionary

Theme: Global AI Adoption & Workforce Displacement Index

Date Range: 2021-Q1 to 2024-Q4

Currency: USD

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- [dim_country]
- [dim_industry]
- [dim_skill_category]
- [dim_date]
- [fact_workforce_ai_index]

[dim_country] Table:

Type: Dimension

Primary Key: country_id

Rows: 30

Description:

One row per country tracked in the index, with macroeconomic and digital readiness attributes used to explain differences in AI adoption and displacement risk.

Columns

Column Name	Data Type	Nullable	Unique	Description
Country_id	Integer	No	Yes (Primary Key)	Unique country identifier
Country_Code	String	No	Yes	ISO 3166-1 alpha-2 country code
Country_name	String	No	Yes	Full country name
Region	String	No	No	Geographic Macro-region
Development_tier	String	No	No	Developed or Emerging economy classification
Gdp_per_capita_usd	Float	No	No	GDP per capita in USD
Digital_infrastructure_score	Float	No	No	0-100 composite score of digital infrastructure quality
Ai_policy_maturity	String	No	No	Nascent, Developing, Established or Advanced
Integer_penetration_pct	Float	No	No	Percentage of population with internet access
Stem_graduates_per_100k	Float	NO	No	Annual STEM graduates per 100,000 population

[dim_industry] Table

Type: Dimension

Primary Key: industry_id

Rows: 25

Description

One row per industry sector, capturing how exposed each sector is to automation and how aggressively it is investing in AI.

Columns

Column name	Data Type	Nullable	Unique	Description
Industry_id	Integer	No	Yes (Primary Key)	Unique industry identifier
Industry_name	String	No	Yes	Industry name
Industry_sector	String	No	No	High-level sector grouping
Automation_susceptibility	Float	No	No	0-100 score of how susceptible the industry is to automation
Avg_ai_investment_pct_revenue	Float	No	No	Average AI investment as percentage of revenue
Ai_adoption_stage	String	No	No	Exploring, Piloting, Scaling, or Mature
Global_workforce_millions	Float	No	No	Estimated global workforce size in millions
Primary_ai_use_case	string	No	No	The most common AI application in this industry

[dim_skill_category] Table:

Type: Dimension

Primary Key: Skill category id

Rows: 8

Description

One row per broad skill category used to classify which kinds of work are most exposed to AI driven displacement or augmentation

Columns

Column name	Data Type	Nullable	Unique	Description
Skill_category_id	Integer	No	Yes (Primary Key)	Unique Skill Category identifier
Skill_category_name	String	No	Yes	Skill Category name
Skill_Category_code	String	No	Yes	Short code for the skill category
Ai_replaceability_score	Float	No	No	0-100 score of how replaceable this skill category is by AI
Ai_augmentation_potential	Float	No	No	0-100 score of how much AI can augment (rather than replace this skill category
Median_reskilling_duration_months	Float	No	No	Typical time required to reskill a worker out of this category
Typical_education_level	String	No	No	Typical education or training level associated with this category
Global_worker_share_pct	Float	No	No	Approximate share of the global workforce in this category (sums to ~100% across all rows)

[dim_date] Table

Type: Dimensions

Primary Key: date_id

Rows: 16

Description

One row per calendar quarter from 2021-Q1 to 2024-Q4, flagging pandemic-affected periods and the onset of the generative AI era (from Q4 2022, following the launch of mainstream generative AI tools)

Columns

Column name	Data Type	Nullable	Unique	Description
Date_id	Integer	No	Yes (Primary Key)	Unique date period identifier
Year	Integer	No	No	Calendar year (2021-2024)
Quarter	Integer	No	No	Calendar Year (1-4)
Year_quarter_label	String	No	No	Human-readable label, e.g. "2023-Q2"
Period_start_date	Date	No	No	First_date_of_quarter

Is_pandemic_period	Boolean	No	No	True if the quarter falls within the COVID-19 disruption period
Generative_ai_era	boolean	No	No	True from 2022-Q4 onward, marking the main stream generative AI period.

Fact_workforce_ai_index

Type: FACT

Primary Key: index_id

Rows: 300

Grain: One Row per country * industry * skill category * quarter observation

Description

Core fact table recording quarterly AI adoption, productivity impact, and workforce displacement metrics for a sampled combination of country, industry, skill category, and time period.

Columns

Column name	Data Type	Nullable	Unique	Description
Index_id	Integer	No	Yes (Primary Key)	Unique Record Identifier
Country_id	Integer	No	No (FK → dim_country)	Country dimension foreign key
Industry_id	Integer	No	No (FK → dim_industry)	Industry dimension foreign key
Skill_category_id	Integer	No	No (FK → dim_skill_category)	Skill category dimension foreign key
Date_id	Integer	No	No (FK → dim_date)	Date dimension foreign key
Ai_adoption_rate	Float	No	No	Percentage (0-100) of the workforce segment actively using AI tools
Productivity_impact_score	Float	No	No	0-10 composite score of AI's productivity impact on this segment.
Displacement_risk_index	Float	No	No	0-10 composite score of displacement risk for this segment
Job_displaced_count	Integer	No	No	Estimated jobs displaced by AI adoption in this segment
Jobs_created_count	Integer	No	No	Estimated jobs created by AI adoption in this segment

Reskilling_investment_usd	Float	No	No	Capital invested in reskilling for this segment in USD
Avg_wage_change_pct	Float	Yes	No	Average wage change (%) attributable to AI adoption; null where wage data is unavailable
AI_tool_usage_hours_per_week	Float	Yes	No	Average weekly hours of AI tool usage per worker; null where usage data is unavailable
Workforce_size	Integer	No	No	Total workforce size for this segment
Data_confidence_score	Float	No	No	0-1 score reflecting the reliability of the underlying survey data

Relationships

Foreign Key Constraints

From Table	From Column	To Table	To Column	Type
Fact_workforce_ai_index	Country_id	Dim_country	Country_id	Many to One
Fact_workforce_ai_index	Industry_id	Dim_industry	Industry_id	Many to One
Fact_workforce_ai_index	Skill_category_id	Dim_skill_category	Skill_category_id	Many to One
Fact_workforce_ai_index	Date_id	Dim_date	Date_id	Many to One

Data Quality & Characteristics

Missing values

Avg_wage_change_pct and ai_tool_usage_hours_per_week each contain a small number (~3%) of nulls in the fact table, representing segments where survey data was unavailable. All other columns are fully populated.

Unique Constraints

Primary key columns are unique with no nulls across all tables.

Temporal Coverage

All period_start_date values fall within 2021-01-01 to 2024-10-01 (start of 2024-Q4)